

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

Group Project Report

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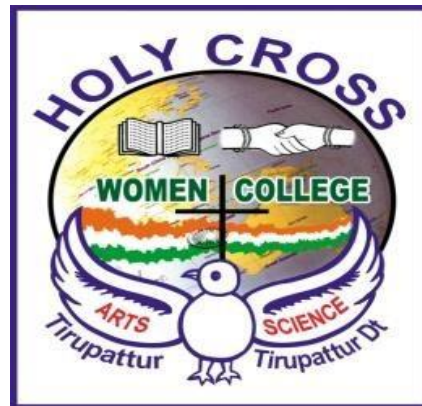
**In Partial Fulfilment for The Award of The Degree
Of**

BACHLER OF COMPUTER SCIENCE

Under the supervision of

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352, HOLY CROSS ARTS AND SCIENCE COLLEGE FOR WOMEN

(Affiliated to Thiruvalluvar University)

DEPARTMENT OF COMPUTER SCIENCE

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Student Management & Course Management System (SECMS)

User Story:

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting within a unified Salesforce environment.

1. INTRODUCTION

1.1 Project Overview:

The **Student Enrollment & Course Management System (SECMS)** is a centralized web-based application designed to automate and streamline the core academic processes of an educational institution. It provides a unified platform for managing student admissions, enrollment, course creation, course registration, attendance, and grade management.

- ☐ Reduce administrative workload and processing time.
- ☐ Minimize data duplication and human errors.
- ☐ Improve data accuracy, consistency, and security.
- ☐ Provide real-time access to academic information for all stakeholders.

The project will be developed using a structured software engineering approach, covering ideation, requirement analysis, system design, planning, and implementation phases, resulting in a scalable and secure solution suitable for colleges and universities.

1.2 Purpose:

The purpose of the Student Enrollment & Course Management System (SECMS) is to provide a single, integrated digital platform that automates and simplifies the key academic and administrative processes related to student enrollment and course management.

- Automate enrollment workflows
- Centralize course and academic data
- Improve efficiency and reduce errors
- Enhance transparency and accessibility
- Support decision-making and reporting
- Ensure security and data integrity

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Question	Answer
Customer	Main Store and Sub Store teams
Cause	Manual and disconnected processes
Impact	Delays, errors, and lack of tracking
Need	Centralized and automated logistics system
Solution	My Logistics – Main Store to Sub Store Management System

1. IDEATION PHASE

1.1 Problem Statement (Customer Problem Statement Template)

S.No	Customer Problem Statement	Description
1	 Who is the customer?	Customers are users who need to transfer materials between the Main Store and Sub Store.
2	 What is the problem?	The material request and transfer process is difficult to manage manually.
3	 Where does the problem occur?	The problem occurs during material request, approval, packing, dispatch, transportation and receiving.
4	 Why is it a problem?	Manual processes can cause delays, errors, missing information and difficulty in tracking materials.
5	 How does it affect the customer?	Customers may not know the current status of their material request or delivery.
6	 Current method used	Requests and material movements may be managed using manual records, phone calls, messages or spreadsheets.
7	 Customer need	Customers need a simple system to raise requests and track material movement in real time.
8	 Expected solution	A centralized logistics management system that connects the Sub Store, Main Store Manager, Packing, Dispatch, Transport and Receiving teams.
9	 Expected benefit	Faster processing, better tracking, reduced errors and improved communication between teams.
10	 Final Problem Statement	There is a need for a centralized logistics management system to efficiently manage and track material transfer between the Main Store and Sub Store from request to delivery.

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

- ☐ Reduce material transfer delays
- ☐ Provide real-time tracking
- ☐ Reduce manual errors
- ☐ Improve communication between teams
- ☐ Provide accurate material information
- ☐ Centralize the entire logistics process
- ☐ Make the system simple and user-friendly

1.1 Empathy Map Canvas



2.3 Brainstorming & Idea Prioritization

A. Brainstorming

- Identify the Problem – Clearly understand the logistics problem.
- Generate Ideas – Encourage team members to suggest different solutions.
- Discuss & Combine – Discuss ideas and combine similar ideas.
- Shortlist Ideas – Select practical and useful ideas.

B. Idea Prioritization

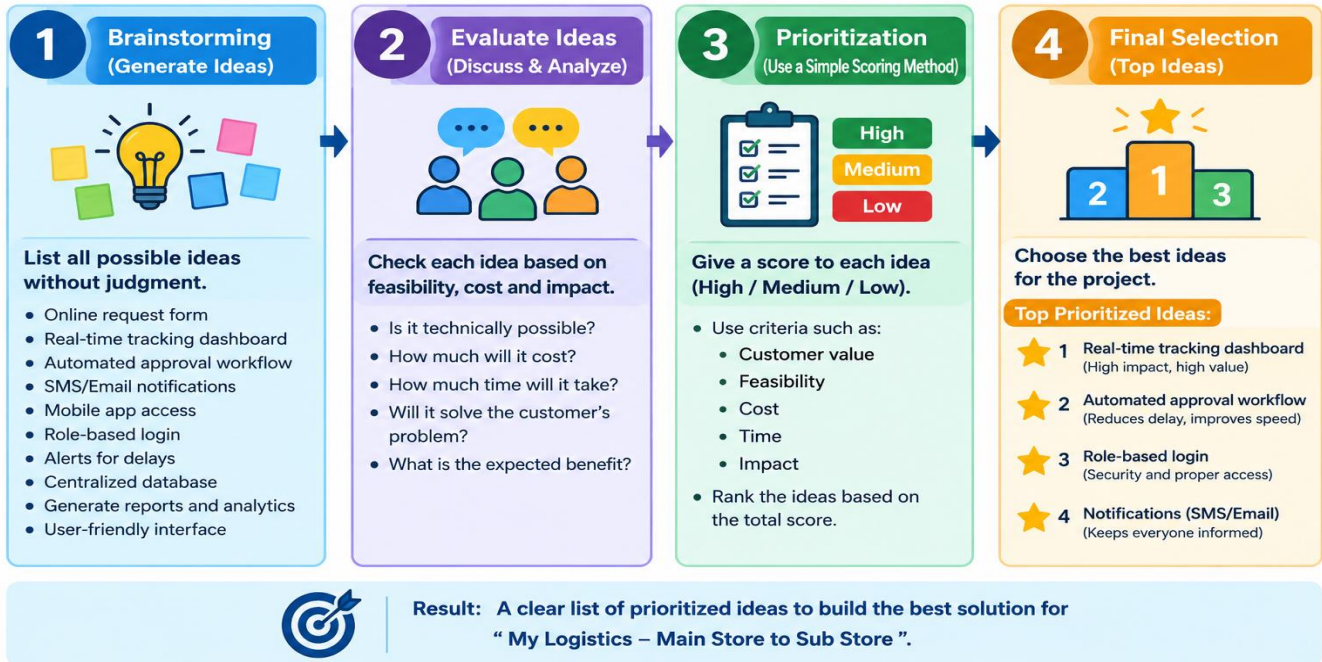
- Define Criteria – Consider impact, feasibility, cost, and effort.
- Score Ideas – Give each idea a score based on the criteria.
- Rank Ideas – Compare scores and arrange ideas by priority.
- Select & Plan – Select high-priority ideas for implementation.

Simple Flow

Problem Identification → Idea Generation → Discussion → Shortlisting → Scoring → Ranking → Final Selection.

2.3 Brainstorming & Idea Prioritization

Generate ideas → Evaluate → Prioritize → Select the best solution

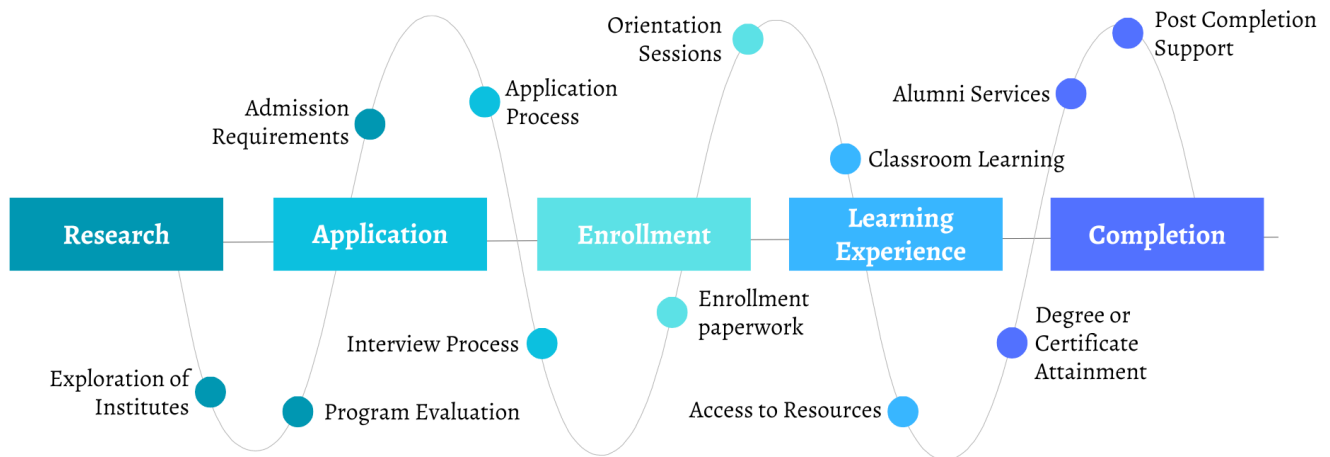


3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

- ☐ Online material request
- ☐ Role-based login
- ☐ Request approval
- ☐ Packing and dispatch management
- ☐ Transport tracking
- ☐ Real-time status updates
- ☐ Receiving confirmation
- ☐ Notification system
- ☐ Centralized database
- ☐ Reports and request history

Customer Journey Map for Education



Non-Functional Requirements

NFR-1: Performance

Meaning: The system should respond quickly when users perform an action.

Example:

When a student logs in or opens the course list, the page should load within a few seconds.

NFR-2: Security

Meaning: Student and course information must be protected from unauthorized access.

Example:

Only the Admin can delete student records, while students can only view and update their own information.

NFR-3: Reliability

Meaning: The system should work correctly and consistently without frequent failures.

Example:

When a student registers for a course, the registration information should be saved correctly.

NFR-4: Usability

Meaning: The system should be simple and easy to use.

Example:

Students should easily find options such as Login, My Courses, Profile, Results, and Logout.

NFR-5: Scalability

Meaning: The system should handle an increasing number of students and courses.

Example:

The system should continue working efficiently when the number of students increases from **500 to 5,000**.

NFR-6: Availability

Meaning: The system should be available whenever users need it.

Example:

Students should be able to access their course details and learning materials at any time, except during.

NFR-7: Backup and Recovery

Meaning: The system should regularly back up important data and recover it after a failure.

Example:

If the database crashes, student records and course information can be restored from the latest.

NFR-8: Compatibility

Meaning: The system should work properly on different devices, browsers, and operating systems.

Example:

Students should be able to use the system on Chrome, Edge, Firefox, mobile phones, tablet.

3.2 Data Flow Diagram

1. External Entities (Sources & Destinations)

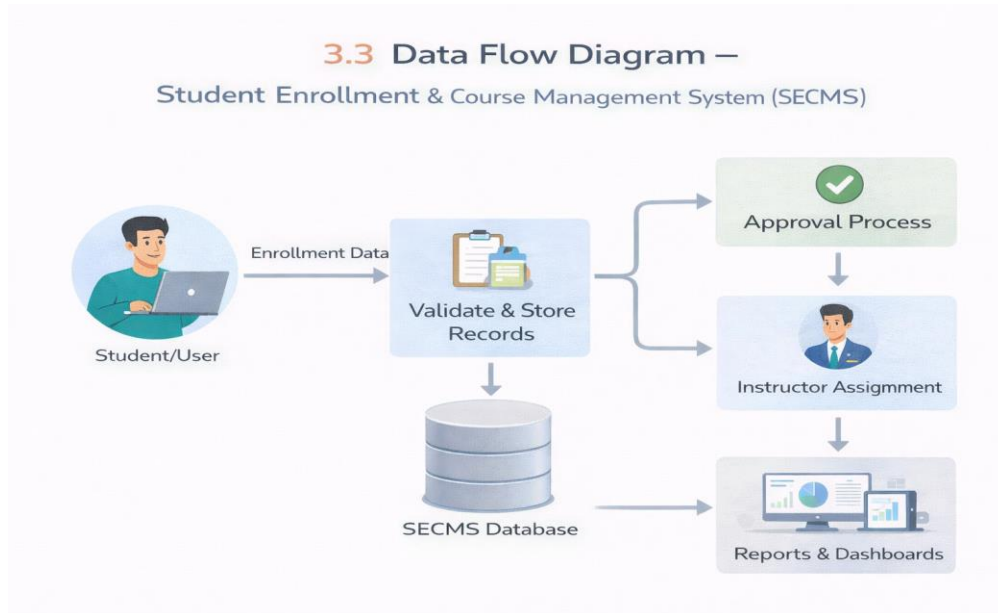
- **Student:** Submits registration, course selection, and views marks/schedules.
- **Faculty:** Submits grades, course materials, and manages attendance.
- **Admin:** Manages course offerings, fee updates, and overall user permissions.

2. DFD Abstraction Levels

Level 0 (Context Diagram): High-level view showing the entire system as a single process node connected to external entities.

Level 1: Breaks the single context process into major sub-processes, primary data stores, and core data flows.

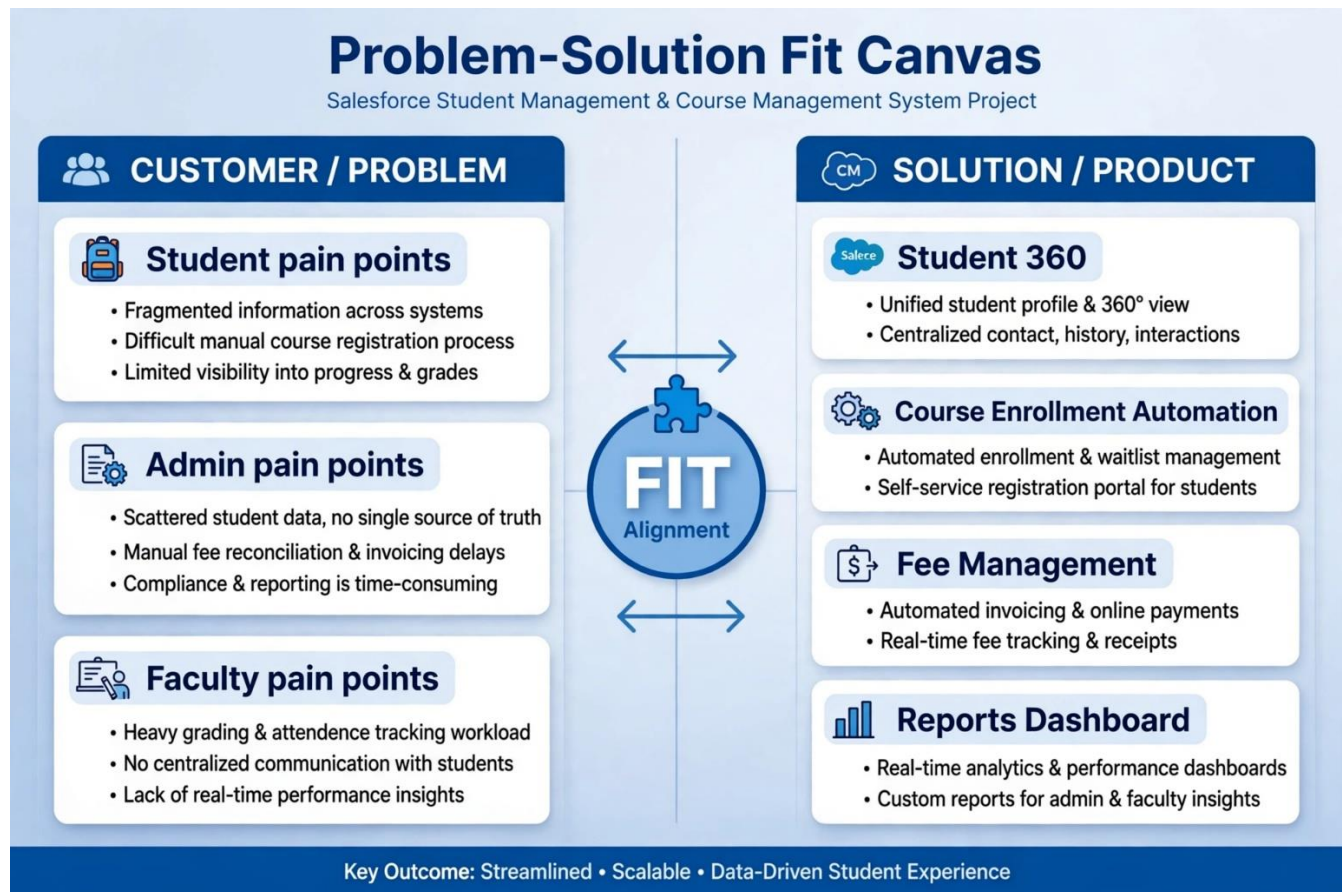
Level 2 (Detailed DFD): Decomposes Level 1 sub-processes further into lower-level functional steps.



4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The purpose of the Problem-Solution Fit is to validate that the proposed Salesforce-based system solves the real pain points of students, admins, and faculty.



4.2 Proposed Solution

The proposed system is a cloud-based Student and Course Management System built on Salesforce Platform to replace manual Excel/Paper-based processes.

Core Objectives:

- Centralize all student data into a single Student 360 view
- Automate course enrolment, fee collection, and grading
- Provide self-service portals for students and faculty
- Enable real-time reporting for management

Key Features Proposed:

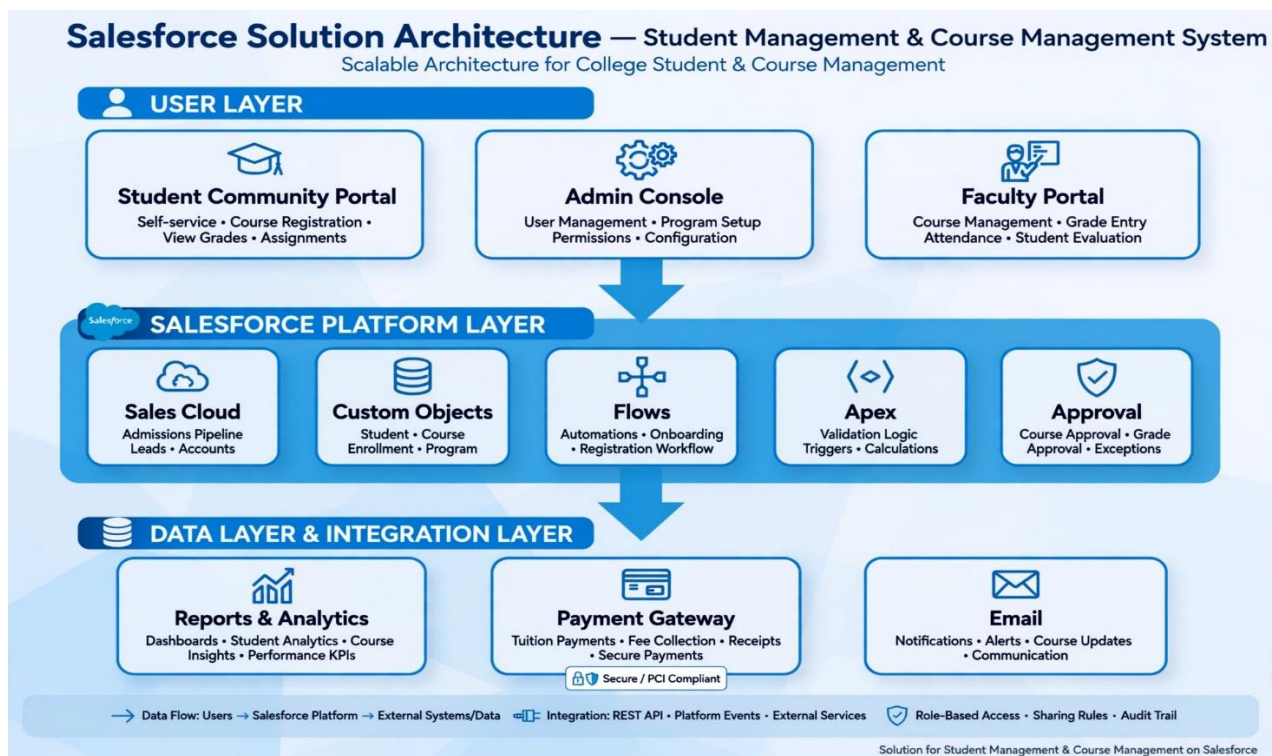
1. Student Management Module:

- Custom Object: Student (fields: Roll No, Contact, Department, Year, Status)
- Auto Student ID generation via Flow
- Duplicate prevention with Matching Rules
- Student lifecycle: Applicant > Enrolled > Alumni

2. Course Management Module:

- Custom Objects: Course, Program, Enrolment
- Fields: Course Code, Credits, Seats, Faculty, Prerequisites
- Validation: Seat availability check, Prerequisite check
- Automated Waitlist management

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

This is the complete list of User Stories for Salesforce Student & Course Management System.

Epic	User Story ID	User Story	Priority	Story Points	Acceptance Criteria
EPIC 1: Student Management	US-01	As an Admin, I want to create a student record with auto ID generation	High	5	Auto ID, Duplicate check, Required fields validation
	US-02	As an Admin, I want to view Student 360 with all enrolments & fees	High	8	Related lists, 360 view pages
	US-03	As a Student, I want to login to portal and view my profile	High	5	Experience Cloud Login
EPIC 2: Course Management	US-04	As an Admin, I want to create Courses with seat limit & prerequisites	High	5	Validation for seats, prereq check
	US-05	As a Student, I want to enrol in courses via self-service portal	High	13	Flow, Seat availability, Waitlist
	US-06	As a System, I want to auto-update seat count on enrolment	Medium	3	Apex Trigger
EPIC 3: Fee Management	US-07	As an Admin, I want to generate fee invoices automatically	High	8	Auto invoice on enrollment
	US-08	As a Student, I want to pay fees online & get receipt	High	8	Payment gateway integration, Email receipt

SPRINT SCHEDULE:

We divided the 6 weeks into 4 Sprints.

Sprint	Duration	Goal	User Stories Included	Deliverable
Sprint 1: Foundation	Week 1 - 1.5	Org Setup & Data Model	US-01, US-04, US-12	Dev Org, Custom Objects: Student, Course, Enrolment, Fee, Roles & Profiles
Sprint 2: Core Development	Week 2 – 3	Automation & Business Logic	US-02, US-06, US-07, US-10	Flows, Apex Triggers (Seat count, GPA), Validation Rules, Approval Process
Sprint 3: Portal & Integration	Week 4 - 5	Community Portal & Payment	US-03, US-05, US-08, US-09	Experience Cloud Student Portal, Screen Flow for enrolment, Email Alerts
Sprint 4: Testing & Deployment	Week 5.5 - 6	Testing, Reports & Go-Live	US-11 + Testing	Test Classes (>75% coverage), Dashboards, UAT, Deployment to Production

6. Project Development Phase:

Project Flow:

Milestone 1: Salesforce Developer Account Creation

Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)

Milestone 3: Tabs Creation

Milestone 4: Lightning App Creation

Milestone 5: Fields Creation

Milestone 6: Create Relationships

Milestone 7: Page Layout Customization

Milestone 8: Validation Rules

Milestone 9: Flows (Automation)

Milestone 10: Approval Process

Milestone 11: Reports

Milestone 12: Dashboards

Milestone 13: Security Setup

Milestone 14: Conclusion

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Develop reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

Milestone 1-Salesforce:

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

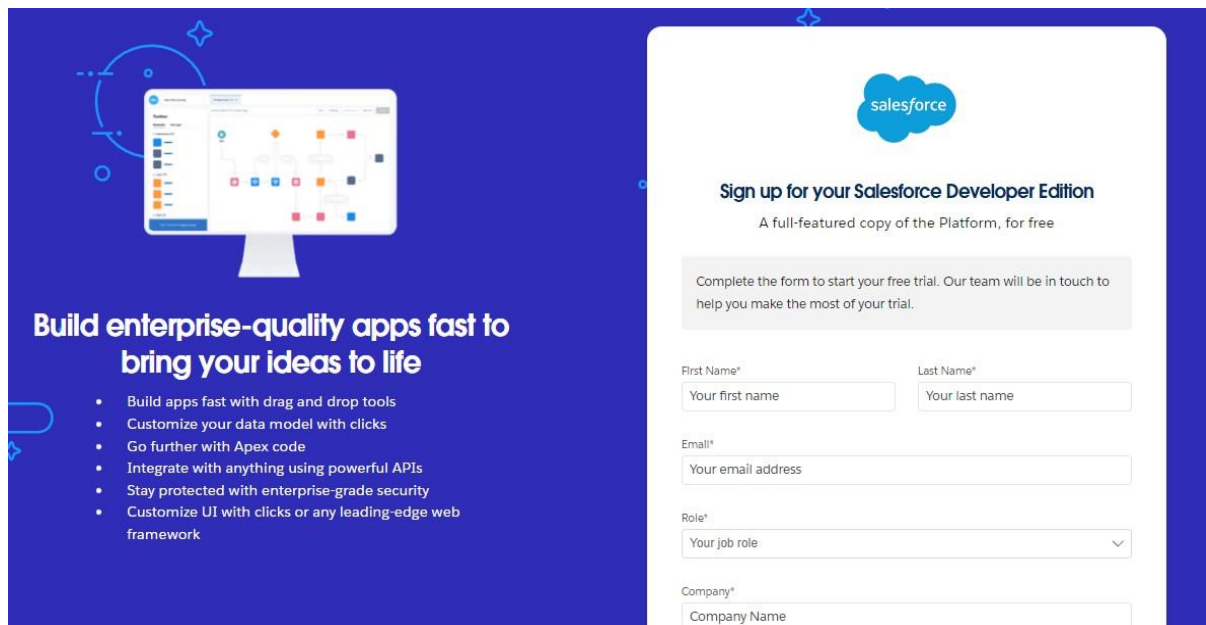
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:

<https://youtu.be/r9EX3IGde5k>

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

Go to <https://developer.salesforce.com/signup> On the sign-up form, enter the following details :



First name & Last name:

Email:

Role: Developer

Company: College Name County: India

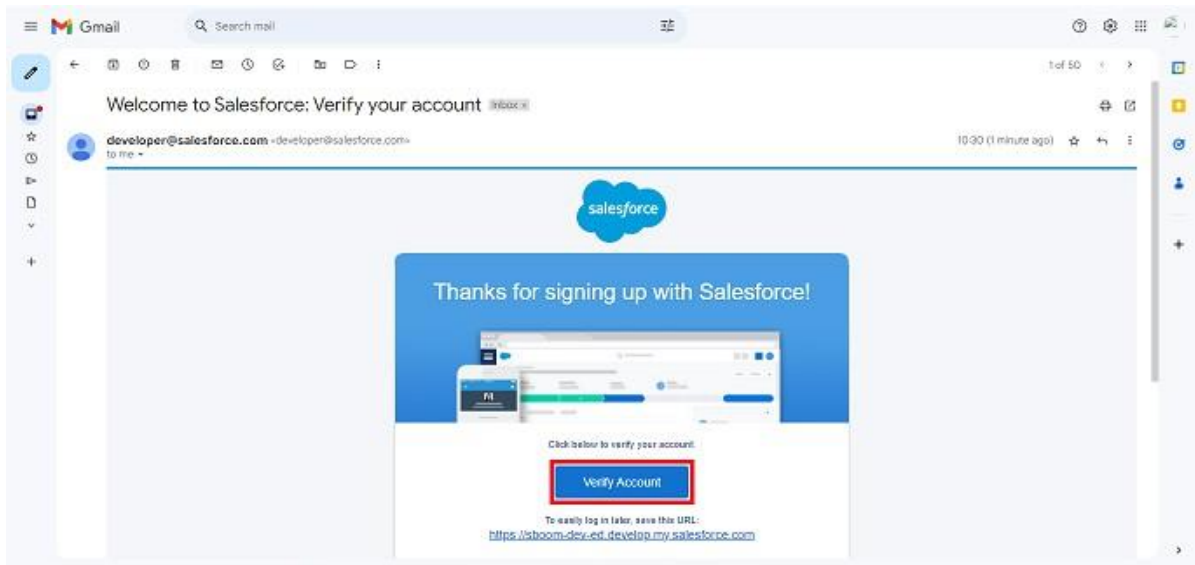
Postal Code: pin code

Username: pavithraarul893@gmail.com

Click on sign me up after filling these.

Activity 2: Account Activation:

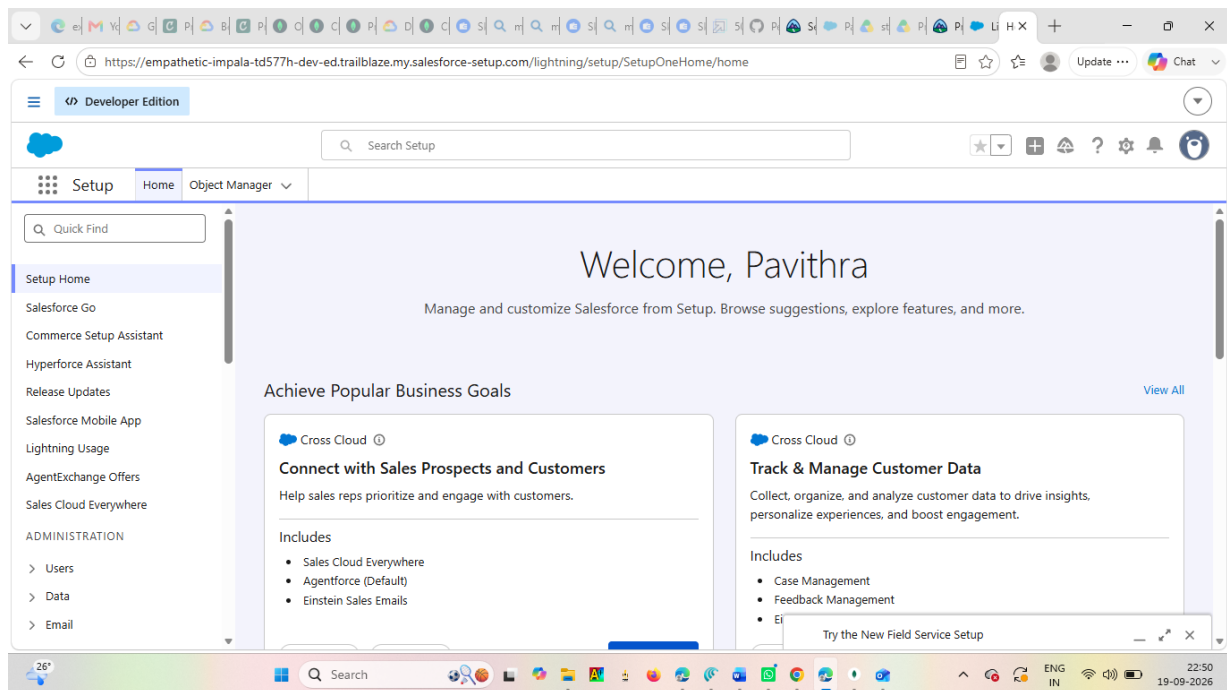
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.

A screenshot of the 'Change Your Password' form. The form has a title 'Change Your Password' and instructions: 'Enter a new password for lead@sb.oom. Make sure to include at least: 8 characters, 1 letter, 1 number'. There are three green checkmarks next to these requirements. The form contains four main sections: 'New Password' with a 'Good' status, 'Confirm New Password' with a 'Match' status, 'Security Question' with a dropdown menu showing 'In what city were you born?', and 'Answer' with the text 'asdfghjkl'. A blue 'Change Password' button is at the bottom, highlighted with a red rectangle.

4. Then you will redirect your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the student object:

Go to the Setup page. Click on

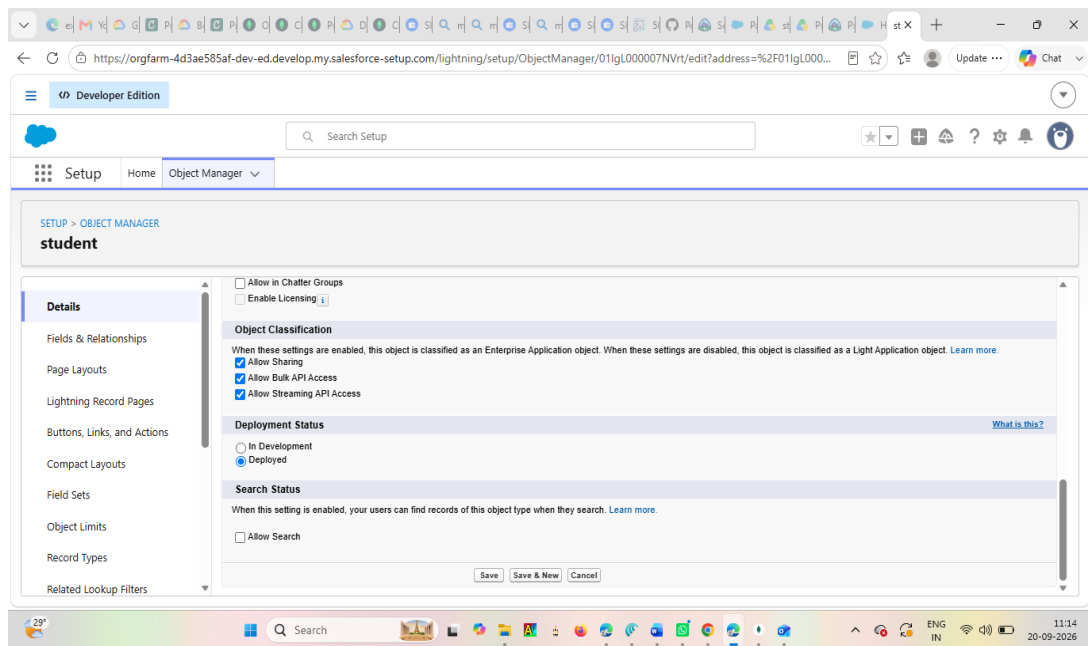
Object Manager.

Click on Create → Custom Object.

Enter the details below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

The screenshot displays the Salesforce Developer Edition interface. The top navigation bar includes the 'Setup' link and the 'Object Manager' dropdown menu. The main content area is titled 'student' and shows the 'Custom Object Definition Edit' page. The left sidebar contains a 'Details' section with links to 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', and 'Related Lookup Filters'. The main content area is divided into two sections: 'Custom Object Information' and 'Custom Object Definition Edit'. The 'Custom Object Information' section includes fields for 'Label' (student), 'Plural Label' (students), 'Object Name' (student), and 'Description'. The 'Custom Object Definition Edit' section includes a 'Save' button, a 'Save & New' button, and a 'Cancel' button. The bottom status bar shows the system time as 11:13 on 20-09-2026.



In the same way Create **Courses, Instructors, and Enrolments objects**

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Table 2: Course Object Fields:

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Table 3: Instructor Object Fields:

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Create Relationships:

Activity 1: Create Lookup from Enrollment to Student:

- Go to Setup → click Object Manager.
- Type Enrollment in Quick Find → click Enrollment object.
- Click Fields & Relationships → click New.
- Select Lookup Relationship → click Next.
- For Related To, select Student c click Next.

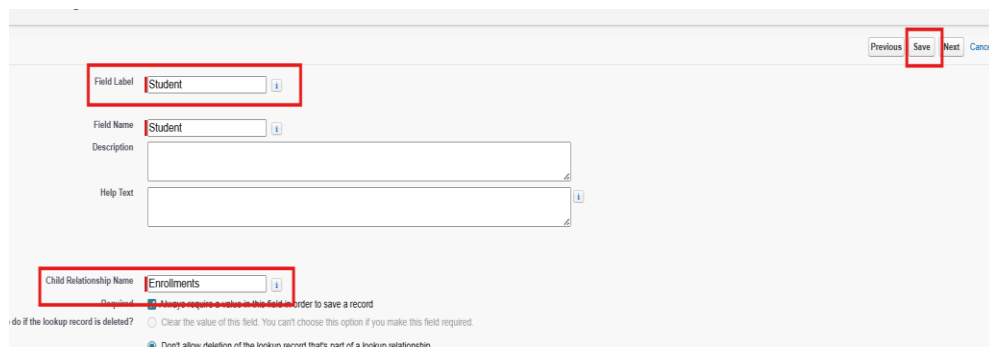
Enter Field Label: Student (API Name: Student_____c).

Click Next through Field-Level Security and Layout options (ensure its visible and added to layout). Click Save.



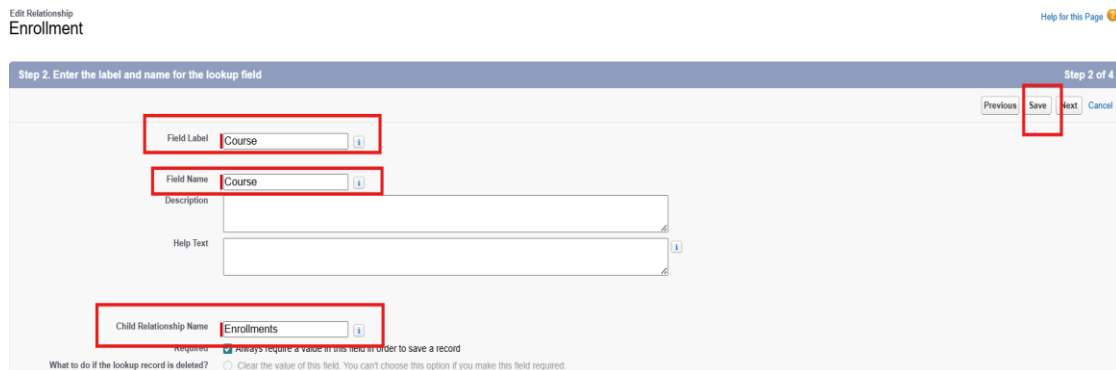
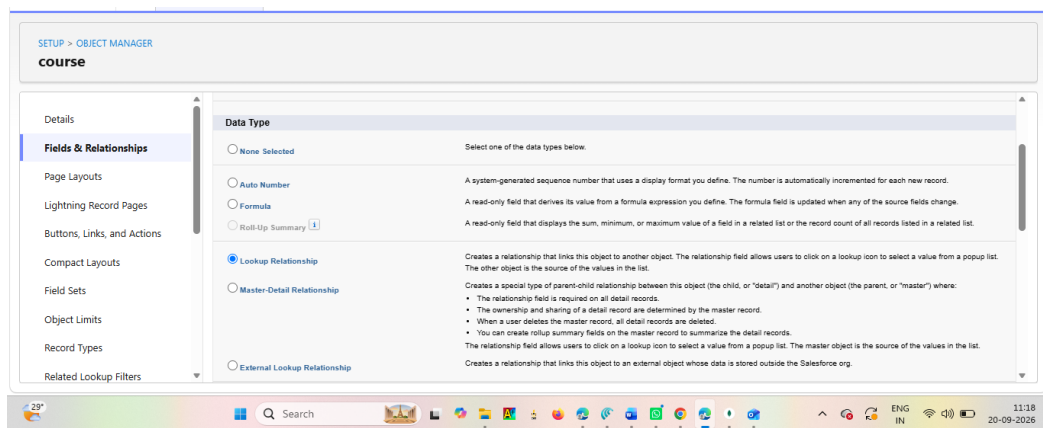
Activity 2: Create Lookup from Enrollment to Course

Purpose: To link each Enrollment to a Course.



Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course_c click Next.
5. Field Label: Course (API Name: Course_____c).
6. Click Next (set visibility and layout) → Save.



Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:

To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor_c Next.
5. Field Label: Instructor (API Name: Instructor_____c).
6. Complete Field-Level Security and Page Layout options → Save

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Specify the type of information that the custom field will contain.

Data Type

Select one of the data types below.

☐ None Selected

☐ Auto Number

☐ Formula

☐ Roll-Up Summary ⓘ

☒ **Lookup Relationship**

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Next Cancel

Enrollment

New Relationship

Help for this Page ⓘ

Step 2. Choose the related object

Previous **Next** Cancel

Select the other object to which this object is related.

Related To **Instructor**

Previous Next Cancel

Step 3. Enter the label and name for the lookup field Step 3 of 6

Previous Next Cancel

Field Label

Field Name

Description

Help Text

Child Relationship Name

Required ☒ Always require a value in this field in order to save a record

What to do if the lookup record is deleted?

☐ Clear the value of this field. You can't choose this option if you make this field required.

☒ Don't allow deletion of the lookup record that's part of a lookup relationship.

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Enrollment

New Relationship

[Help for this Page](#)

Step 6. Add custom related lists Step 6 of 6

Previous Save & New Save Cancel

Field Label Instructor

Data Type Lookup

Field Name Instructor

Description

Specify the title that the related list will have in all of the layouts associated with the parent.

Related List Label

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout. To change the values that appear, you will need to customize the Record Types.

Add Related List	Page Layout Name
☒	Instructor Layout

☒ Append related list to users' existing personal customizations

Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

Purpose:

To ensure students use only allowed institutional email addresses. Steps:

Go to Setup → Object Manager → Student_____c.

Click Validation Rules → New.

Rule Name: Student_Email_Domain_Validation. Error Condition

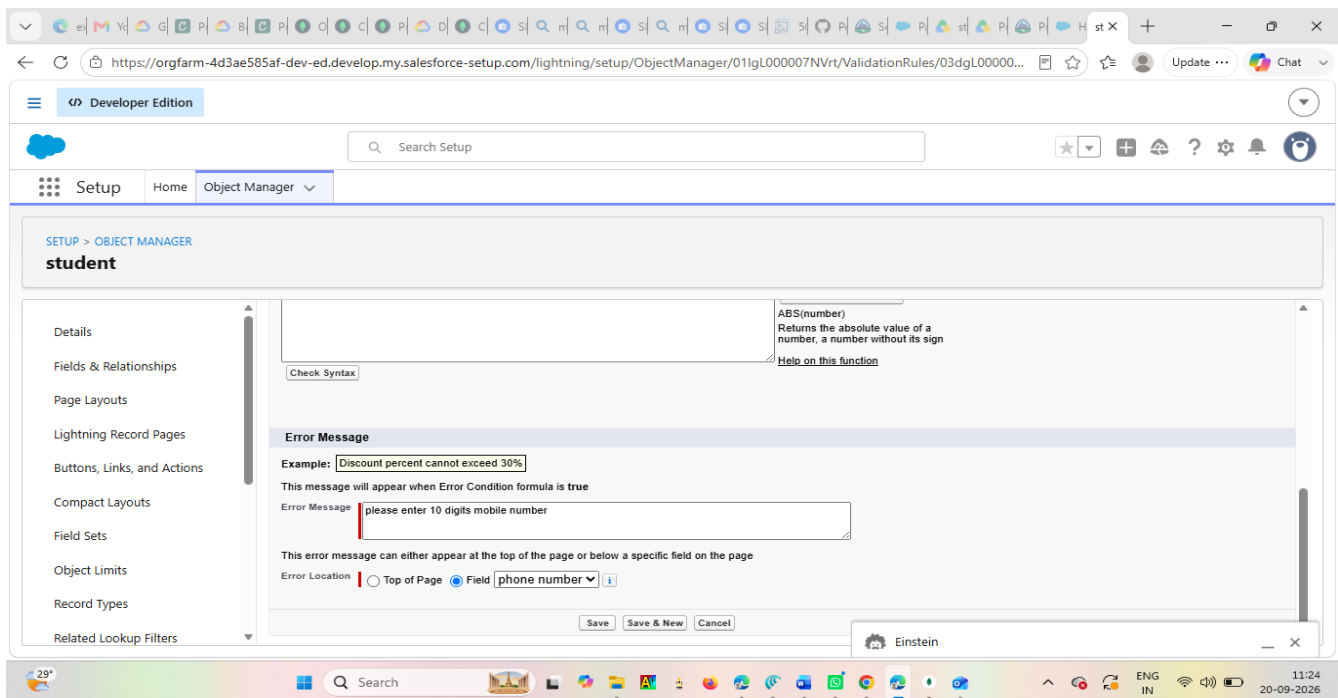
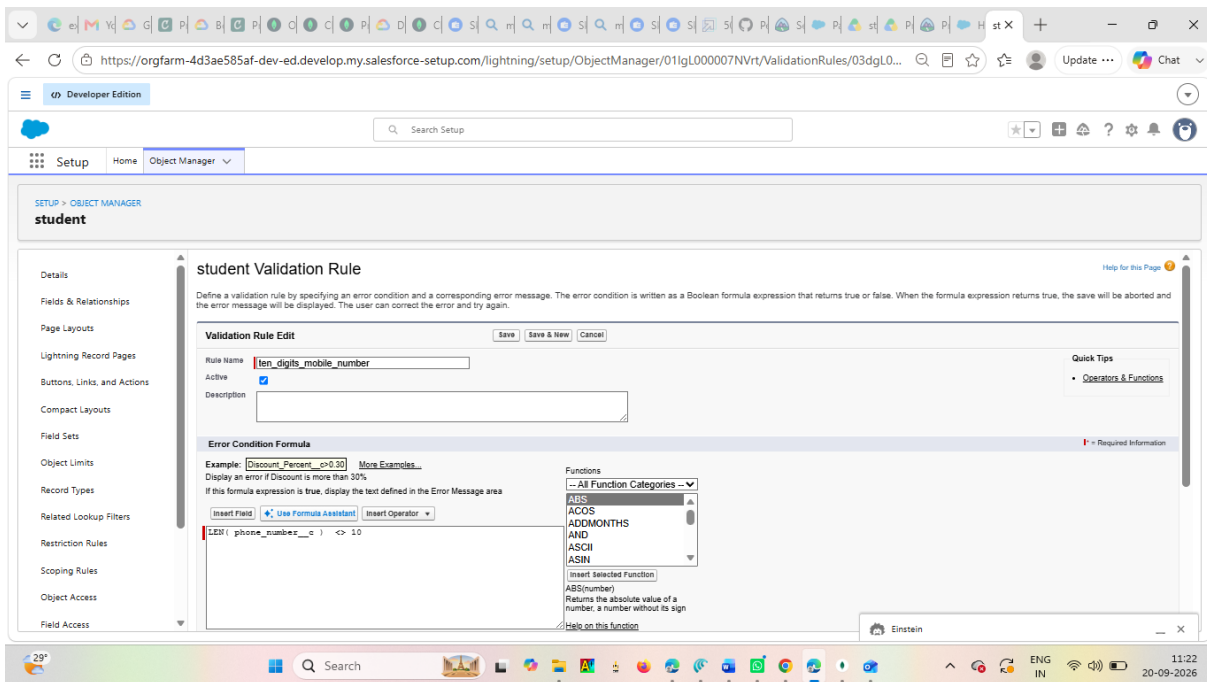
Formula: (example for student.edu)

NOT (CONTAINS(Email_____c, "@student.edu"))

Error Message: Email must be a valid student institutional email. Error Location:

Field → Email_____c.

Click Save.



Activity 2: Fees Must Be Paid Before Approval

Purpose:

Prevent Enrollment from being approved until fees are paid. Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:

```
AND(  
    ISPICKVAL(Enrollment_Status_____c, "Approved"),  
    Fees_Paid_____c = FALSE  
)
```

5. Error Message: Fees must be paid before approving an enrollment.

6. Error Location: Top of Page (or field as needed). Click Save.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a navigation menu with the following items: Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Triggers, Flow Triggers, and Validation Rules. The 'Validation Rules' item is highlighted. The main content area is titled 'Enrollment' and shows the configuration for a new validation rule. The 'Rule Name' field is set to 'Fees_Must_Be_Paid_Before_Approval'. The 'Active' checkbox is checked. The 'Error Condition Formula' section displays the formula:

```
AND(  
    ISPICKVAL(Enrollment_Status_____c, "Approved"),  
    Fees_Paid_____c = FALSE  
)
```

. The 'Check Syntax' button is visible. A 'Functions' panel on the right lists various functions, including ABS, ACOS, ADDMONTHS, AND, ASCII, and ASIN. The 'AND' function is selected.

Error Message

Example: Discount percent cannot exceed 30%

This message will appear when Error Condition formula is **true**

Error Message: Fees must be paid before approving an enrollment

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ☒ Top of Page ☐ Field [i](#)

Save **Save & New** **Cancel**

Milestone 6 – Approval Process

Activity 1: Create Approval Process for Enrollment Requests

Purpose:

Route Enrollment requests to Training Manager for approval or rejection.

Steps:

1. Go to Setup → Approval Processes → select Enrollment_____c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment_Status_____c = "Requested".
5. Save and configure steps.

Setup Home Object Manager

approval

Data

Mass Transfer Approval Requests

Feature Settings

Approval Settings

Process Automation

Approval Processes

Didn't find what you're looking for? Try using Global Search.

SETUP

Approval Processes

Open Approvals App

Approvals are complex business processes that require information gathering and planning before implementing. It is recommended that you follow the instructions below before getting started.

1. Read the help topic
2. View the checklist
3. Create a custom user hierarchical relationship field
4. Create email templates
5. Create an approval process using either the Jump Start or Standard Wizard
6. Add Approval History Related List to all page layouts
7. Activate the process to deploy to your users

Manage Approval Processes For Enrollment

A listing of both active and inactive approval processes for **Enrollments** is displayed below. To create a new approval process, click Create New Approval Process then select Use Jump Start Wizard to set up your short steps. Or, select Use Standard Wizard to configure all approval options.

Create New Approval Process

Use Jump Start Wizard

Use Standard Setup Wizard

Reorder

Approval Process Edit

Approval Enrollment Request

Step 1. Enter Name and Description

Enter a name and description for your new approval process.

Enter Name and Description

Process Name	Approval Enrollment Request
Unique Name	Approval_Enrollment_Request
Description	Approval process for Enrollment requests — sends to Training Manager

Step 2. Specify Entry Criteria

Previous Save **Next**

If only certain types of records should enter this approval process, enter that criteria below. For example, only expense reports from employees at headquarters should use this approval process.

Specify Entry Criteria

Use this approval process if the following criteria are met

Field	Operator	Value	
Enrollment: Enrollment Status	equals	Requested	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		

6.Add Approval Step:

7.Approver: specific Training Manager user.

SETUP
Approval Processes

Approval Step Edit
Training Manager [Help for this Page](#)

Step 1. Enter Name and Description Step 1 of 3

Save Next Cancel

Enter a name, description, and step number for your new approval step.

Enter Name and Description ! = Required Information

Approval Process Name: Approval Enrollment Request

Name: Training Manager

Unique Name: Training_Manager

Description:

Save Next Cancel

1. Under Final Approval Actions:
2. Add Field Update to set Enrollment_Status_____c = "Approved".

SETUP
Field Updates

Define the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the fields are shown only for the type that you select.

Field Update Edit Save Save & New Cancel

Identification

Name: Status to Approved

Unique Name: Status_to_Aproved

Description:

Object: Enrollment

Field to Update: Enrollment: Enrollment Status

Field Data Type: Picklist

Re-evaluate Workflow Rules after Field Change: ☐

Specify New Field Value

Picklist Options

☐ The value above the current one

☐ The value below the current one

☒ A specific value: Approved

Save Save & New Cancel

3. Under Final Rejection Actions:
4. Add Field Update to set Enrollment_Status_____c = "Rejected".

Field Update Edit Save Save & New Cancel

Identification

Name **Set Status to Rejected**

Unique Name **Set_Status_to_Rejected** ⓘ

Description

Object Enrollment

Field to Update Enrollment: Enrollment Status

Field Data Type Picklist

Re-evaluate Workflow Rules after Field Change ☐ ⓘ

Specify New Field Value

Picklist Options

☐ The value above the current one

☐ The value below the current one

☒ A specific value **Rejected** ▼

5. Add Email Alert to send rejection email to students.

Email Alert Edit Save Save & New Cancel

Edit Email Alert

Description **Enrollment Rejected**

Unique Name **Enrollment_Rejected** ⓘ

Object **Enrollment**

Email Template **Enrollment Rejected** ⓘ

Protected Component ☐

Recipient Type Search: **User** for: Find

Recipients

Available Recipients

User: Integration User ⓘ

User: OrgFarm EPIC

User: Security User

Selected Recipients

User: sai kumar A ⓘ

Add

Remove

You can enter up to five (5) email addresses to be notified.

Additional Emails **geespam198@gmail.com**

From Email Address **Current User's email address** ▼

☐ Make this address the default From email address for this object's email alerts. ⓘ

Save Save & New Cancel

6. Activate the approval process.

7. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

In the Quick Find box, type Tabs.

Under Custom Object Tabs, click New.

Select Student from the object list.

1. Choose a tab style (icon).
2. Click Next, assign it to all profiles, and then click Save.

The screenshot shows the Salesforce Setup interface. On the left sidebar, the 'Setup' menu is open, and 'Tabs' is highlighted under 'Custom Object Tabs'. The main content area is titled 'Edit Custom Object Tab: Students'. It contains a 'Custom Tab Definition Edit' section with the following fields:

- Tab Label:** Students
- Object:** Student
- Tab Style:** Airplane (with a search icon)

Below these fields, there is a dropdown for 'Splash Page Custom Link' set to '--None--'. A text area for 'Description' is also present. At the bottom, there are 'Save' and 'Cancel' buttons.

Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.

Follow the same steps as mentioned in Activity -1

Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box. Click New Lightning App.

Provide the app details:

App Name: Student Enrollment & Course Management Developer Name auto-fills

Description: Optional, but can describe the purpose of the app Click Next and choose the following settings:

App Branding (optional): Add logo or select color theme Navigation Style: Choose Standard Navigation

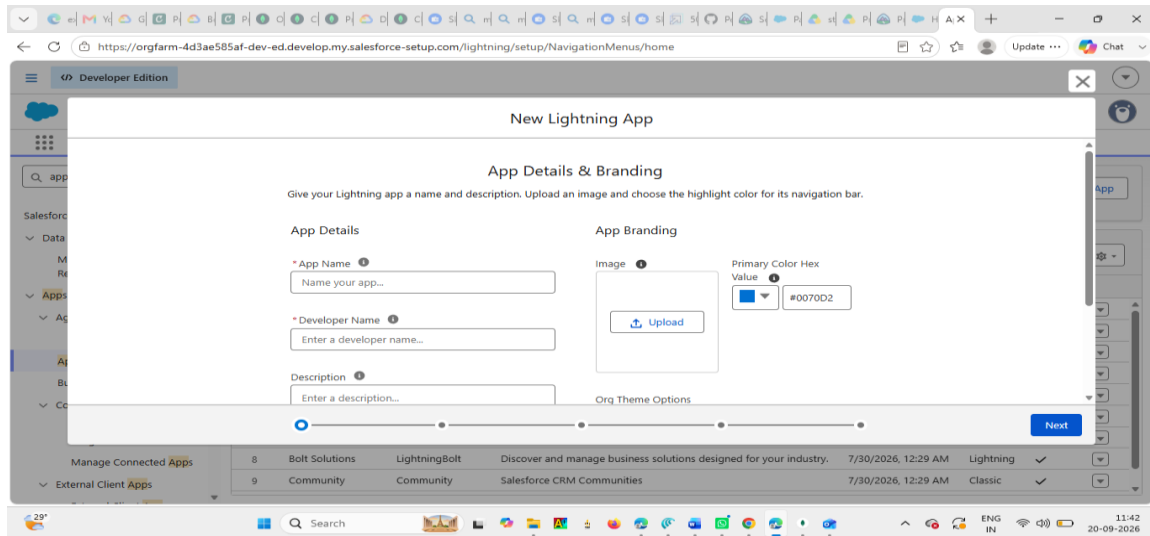
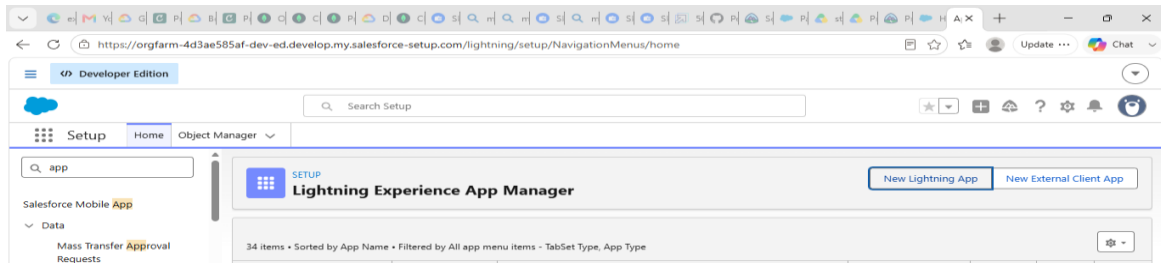
Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).

Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

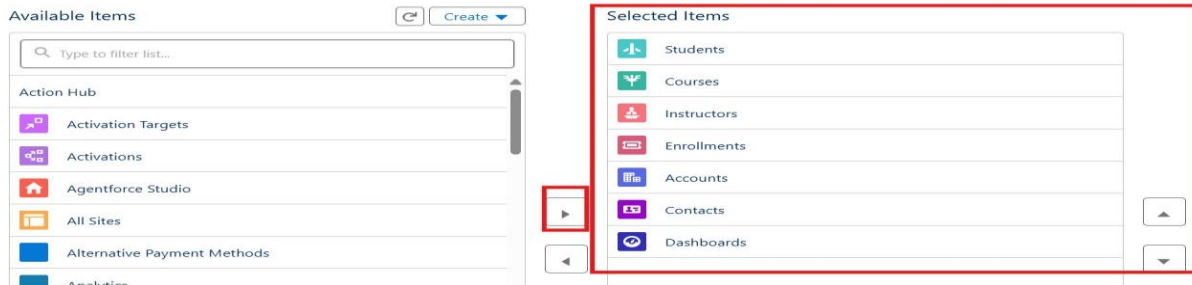
Click Next and review the app settings.

Finally, click Save & Finish.



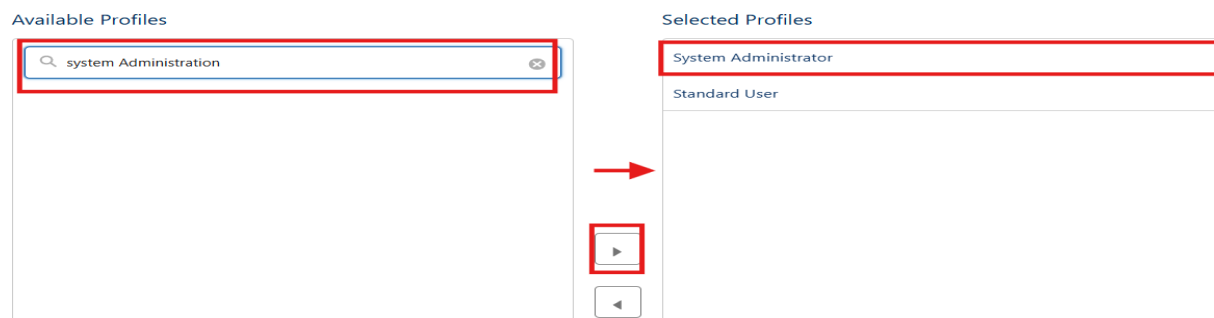
Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.



User Profiles

Choose the user profiles that can access this app.



Milestone 9 – Page Layouts and Record:

Activity 1: Create Record Types for Enrollment:

Purpose:

Separating New Enrollment vs Re-Enrollment processes. Steps:

1. Go to Setup → Object Manager → Enrollment_____c.

2. Click Record Types → New.

3. For the first record type:

Existing Record Type: Master

Record Type Label: New Enrollment

Enable for required profiles (Admin, Enrollment Officer).

Select the default Page Layout → Save & New.

4. For the second record type:

Record Type Label: Re-Enrollment

Enable for same profiles.

Select the default Page Layout → Save.

The screenshot shows the Salesforce Setup page for Enrollment Record Types. The left sidebar contains a list of options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types (highlighted with a red box), and Related Lookup Filters. The main content area is titled 'Edit Record Type Re-Enrollment' and includes a prompt to 'Enter a new name for the selected record type and click Save.' Below this, the 'Record Type' section contains the following fields: 'Record Type Label' (highlighted with a red box and containing 'Re-Enrollment'), 'Record Type Name' (highlighted with a red box and containing 'RE_ENROLLMENT'), 'Namespace Prefix' (empty), 'Description' (containing 'Used when a student re-enrolls in a course'), and 'Active' (checked). At the bottom right of the form are 'Save' and 'Cancel' buttons, with the 'Save' button highlighted by a red box.

Activity 2: Separate Page Layo Creations for Each Enrollment Record Type

The screenshot shows the Salesforce Setup page for New Enrollment Record Type. The left sidebar contains a list of options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types (highlighted with a red box), and Related Lookup Filters. The main content area is titled 'Edit Record Type New Enrollment' and includes a prompt to 'Enter a new name for the selected record type and click Save.' Below this, the 'Record Type' section contains the following fields: 'Record Type Label' (highlighted with a red box and containing 'New Enrollment'), 'Record Type Name' (highlighted with a red box and containing 'New_Enrollment'), 'Namespace Prefix' (empty), 'Description' (containing 'Used for first-time enrollments'), and 'Active' (checked). At the bottom right of the form are 'Save' and 'Cancel' buttons, with the 'Save' button highlighted by a red box.

Purpose:

To show different fields for New Enrollment vs Re-Enrollment. Steps:

1. Enrollment_____c object → click Page Layouts.

2. Click New to create a layout for New Enrollment:

Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.

Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.

Remove unnecessary re-enrollment fields if any.

Click Save.

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Create New Page Layout

As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard

Existing Page Layout: Enrollment Layout

Page Layout Name: New Enrollment Layout

Save Cancel

3. Again, click New to create Re-Enrollment Layout:

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

Quick Find Field Name

Section	Course Fee	Enrollment Status	Owner
Blank Space	Created By	Fees Paid	Record Type
Comments	Enrollment Name	Instructor	Student
Course	Enrollment Date	Last Modified By	

Information (Header visible on edit only)

Enrollment Name	Sample Text
Student	Sample Text
Course	Sample Text
Instructor	Sample Text
Enrollment Date	12/22/2025
Enrollment Status	Sample Text
Fees Paid	✓
Comments	Sample Text
Course Fee	\$123.45

Clone existing layout → name its Re-Enrollment Layout → Save.

- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar shows the 'Enrollment' object with various tabs: Details, Fields & Relationships, Page Layouts (highlighted), Lightning Record Pages, Buttons, Links, and Actions, and Compact Layouts.

The main content area displays the 'Create New Page Layout' dialog. It includes a message: 'As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard sections whose i'. Below this, there are two input fields: 'Existing Page Layout' with a dropdown menu showing 'Enrollment Layout', and 'Page Layout Name' with the text 'Re-Enrollment Layout'. At the bottom right of the dialog are 'Save' and 'Cancel' buttons.

Below the dialog, the 'Page Layout Assignment' table is visible. The table has columns for 'Fields', 'Section', 'Field Name', and 'Record Type'. The 'Fields' column lists various fields like 'Enrollment Name', 'Student', 'Course', 'Enrollment Status', 'Fees Paid', 'Enrollment Date', 'Comments', 'Course Fee', and 'Instructor'. The 'Section' column lists 'Section' and 'Blank Space'. The 'Field Name' column lists 'Course Fee', 'Created By', 'Enrollment Name', 'Enrollment Date', 'Enrollment Status', 'Fees Paid', 'Instructor', 'Last Modified By', 'Owner', and 'Record Type'. The 'Record Type' column lists 'Student'.

The 'Page Layouts' tab in the left sidebar is highlighted with a red box. The 'Save' button in the 'Create New Page Layout' dialog is also highlighted with a red box. The 'Page Layout Assignment' table is also highlighted with a red box.

4. Click Page Layout Assignment → Edit Assignment.

5. For each profile and record type:

New Enrollment → use New Enrollment Layout.

6. Re-Enrollment → use Re-Enrollment Layout. Click Save.

Activity 3: Enrollment Related Lists to Parent Objects:

Purpose:

To see all Enrollments from Student, Course, and Instructor records.

a) Student → Enrollment Related List

Go to Setup → Object Manager → Student_____c.

Click Page Layouts → open the main layout.

In the palette, select Related Lists.

Drag Enrollments related list onto the layout. Customize columns if needed → Save.

b) Course → Enrollment Related List

Go to Object Manager → Course_____c → Page Layouts. Open layout → add Enrollments related list.

Save.

c) Instructor → Enrollment Related List

Go to Object Manager → Instructor_____c → Page Layouts.

Open layout → add Enrollments related list.

Save.

Milestone 10 – Automation with Flow

Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Set Enrollment Date_____c to today when Enrollment is created.

Steps:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment_____c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

* Object

Enrollment

Configure Trigger

Trigger the Flow When:

- ☒ A record is created
- ☐ A record is updated
- ☐ A record is created or updated
- ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of records that trigger the flow to conserve your org's resources.

Condition Requirements

None

6. Add Update Records element:

- Label: Set Enrollment Date.
- Use record that triggered the flow.
- Field: Enrollment Date_____c → Value: \$Flow.CurrentDate.

7. Save as Set Enrollment Date → Activate.

Update Records

* Label

Set Enrollment Date

* API Name

Set_Enrollment_Date

Description

* How to Find Records to Update and Set Their Values

☒ Use the enrollment record that triggered the flow

☐ Update records related to the enrollment record that triggered the flow

☐ Use the IDs and all field values from a record or record collection

☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record

None—Always Update Record

Set Field Values for the Enrollment Record

Field

Enrollment_Date

Value

Running Flow Interview > CurrentDate

Activity 2: Send Email When Enrollment Approved

Purpose:

When Enrollment Status_____c = Approved, send email, alert admin, and set Student to Active.

Steps (high level for document):

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment_____c, Trigger on update.
4. Condition: Enrollment Status_____c Equals "Approved".

Configure Start

Enrollment

Configure Trigger

Trigger the Flow When:

- ☐ A record is created
- ☐ A record is updated
- ☒ A record is created or updated
- ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Enrollment Status	Equals	Approved

5. Add Get Records to fetch related Student_____c.

Get Records

* Label

* API Name ⓘ

Description

Get Records of This Object

* Data Source

☐ Data Cloud Object
 ☒ Salesforce Object

* Object

Filter Student Records

Condition Requirements

All Conditions Are Met (AND) ▼

Field

Operator

Value

⋮

Status

×

Equals

▼

Aa

Active

×

+ Add Condition

6. Add Action (Email Alert) to send email to students.

Enrollment Approved EmailAlert

* Label

* API Name ⓘ

Description

Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

Aa * Record ID ⓘ

Aa

Triggering Enrolement_c > Record ID

×

7. Add Action to notify Admin if needed.

8. Add Update Records to set Student_____c.Status_____c = "Active".

Update Records

* Label: Set Student Status Active

* API Name: Set_Student_Status_Active

Description:

*** How to Find Records to Update and Set Their Values**

- ☒ Use the enrollment record that triggered the flow
- ☐ Update records related to the enrollment record that triggered the flow
- ☐ Use the IDs and all field values from a record or record collection
- ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record: None—Always Update Record

Set Field Values for the Enrollment Record

Field	Value
Comments	True

9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

Purpose:

Assign Instructor automatically based on Course category. Rules:

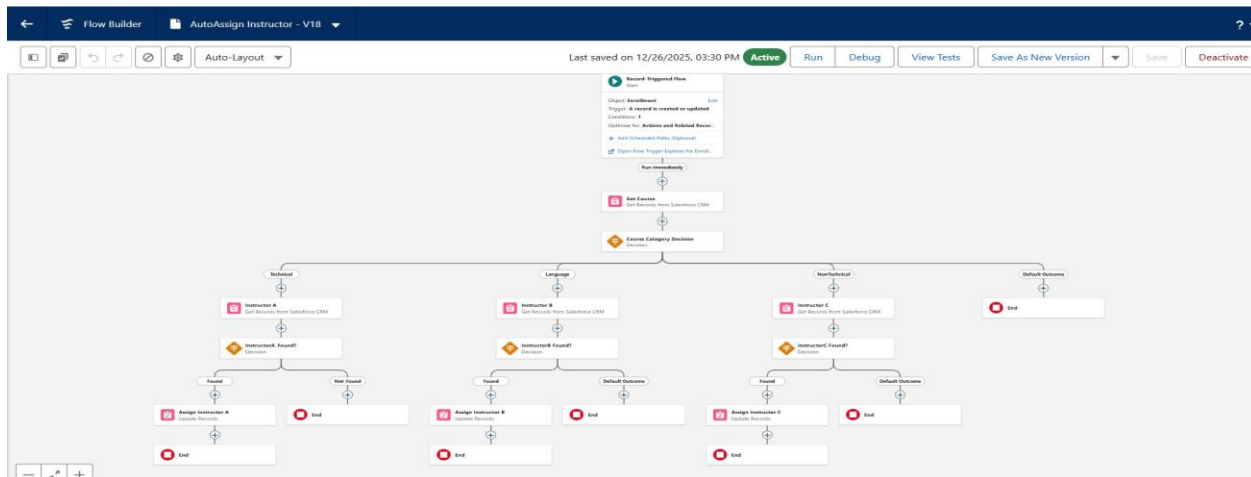
Technical → Instructor A Language

→ Instructor B Non-Technical →

Instructor C

Steps:

1. Prepare Instructor_____c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment_____c (created/updated).
3. Add Get Records to fetch related Course_____c.
4. Add Decision element on Course_____r.Category_____c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor_____c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



Activity 4: Auto-Create Task for Follow-Up

Purpose:

When Enrollment is “Requested”, create a follow-up task for Admin.

Steps:

1. Create Record-Triggered Flow on Enrollment_____c.
2. Trigger: Record created or updated; condition Enrollment_Status_____c
= "Requested".

Configure Start ×

* Object
Enrollment

Configure Trigger

Trigger the Flow When:

☐ A record is created
☐ A record is updated
☒ A record is created or updated
☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements
All Conditions Are Met (AND) ▼

Field	Operator	Value
Enrollment Status	Equals	Requested

3. Add Get Records to fetch Admin user (by Username / Profile).

4. Add Formula (Date): TODAY () + 2 for due date.

Get Records ×

Get Admin User Get_Admin_User

Description

Get Records of This Object

* Data Source

☐ Data Cloud Object
☒ Salesforce Object

* Object
User

Filter User Records

Condition Requirements
All Conditions Are Met (AND) ▼

Field	Operator	Value
Username	Equals	geespam198430@agentforce.com

+ Add Condition

5. Add Create Records for Task:

6. Subject: Follow up on enrollment request.

7. Owner Id: Admin user Id.

8. What Id: Enrollment Id.

9. ActivityDate: formula Today + 2.

Create Records

×

Create a Record of This Object

* Object

Task

Set Field Values for the Task

Field

Due Date Only

←

Value

DueDateTodayPlus2

✕

🗑️

Field

Description

←

Value

Auto-created follow-up for enrollment request

🔍

🗑️

Field

Priority

←

Value

Normal

✕

🗑️

Field

Status

←

Value

Not Started

✕

🗑️

Field

Subject

←

Value

Follow up on enrollment request

🔍

🗑️

Field

Related To ID

←

Value

Triggering Enrolement_c > Record ID

✕

🗑️

10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

Activity 1: Create “Students by Status” Report

- Go to **Reports** → **New Report**.
- Report Type: **Students**.
- Group rows by **Status_____c**.
- Add columns as needed (Name, Email, Phone).
- Save as **Students by Status**.

REPORT 1

Students by Status

Students

Save & Run

Save

Close

Run

Update Preview Automatically

Previewing a limited number of records. Run the report to see everything.

Record Count

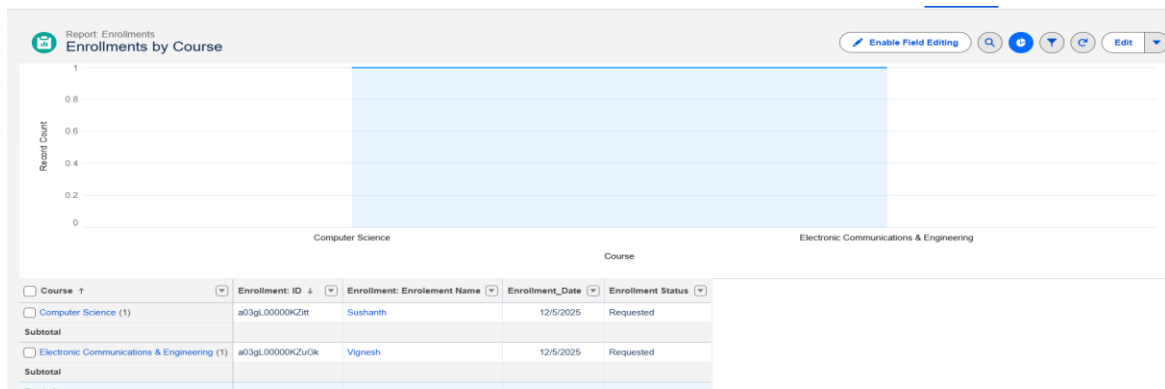
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6 2.8 3 3.2 3.4 3.6 3.8 4

Status

Active

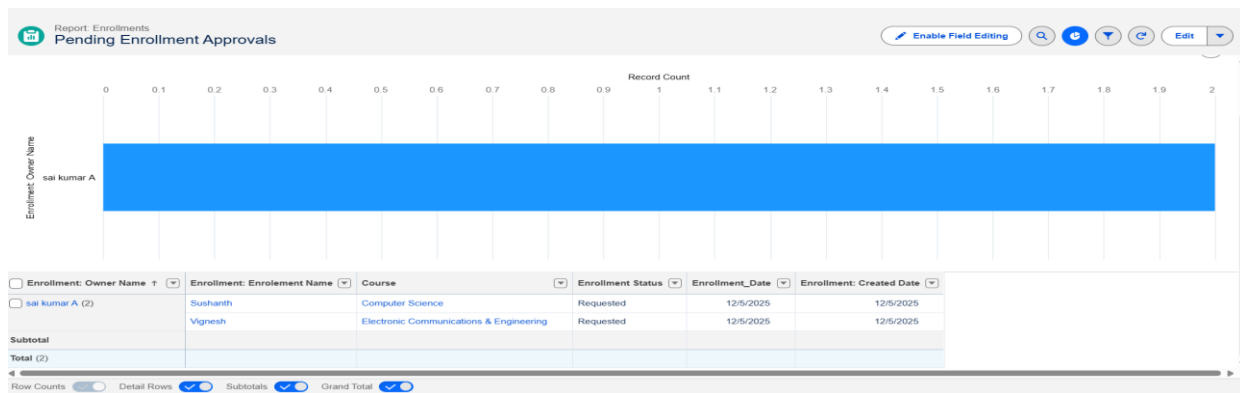
4

Status	Student: Student Name	Last Name	Phone	Email
Active (4)	Abhishek	Jakkula	12345625	abhishek@student.edu
	Vignesh	Doodh	12345678	vignesh@student.edu
	Swaroop	M	212154231	swroop@student.edu
	Sushanth	jalgama	9000000015	sushanth@student.edu
Subtotal				
Total (4)				



Activity 3: Create “Pending Enrollment Approvals” Report

- 1 . New Report → **Enrollments** report type.
- 2 . Filter: **Enrollment_Status**_____c = "Requested".
- 3 . Add columns (Student, Course, Owner, Created Date).
- 4 . Save as **Pending Enrollment Approvals**.

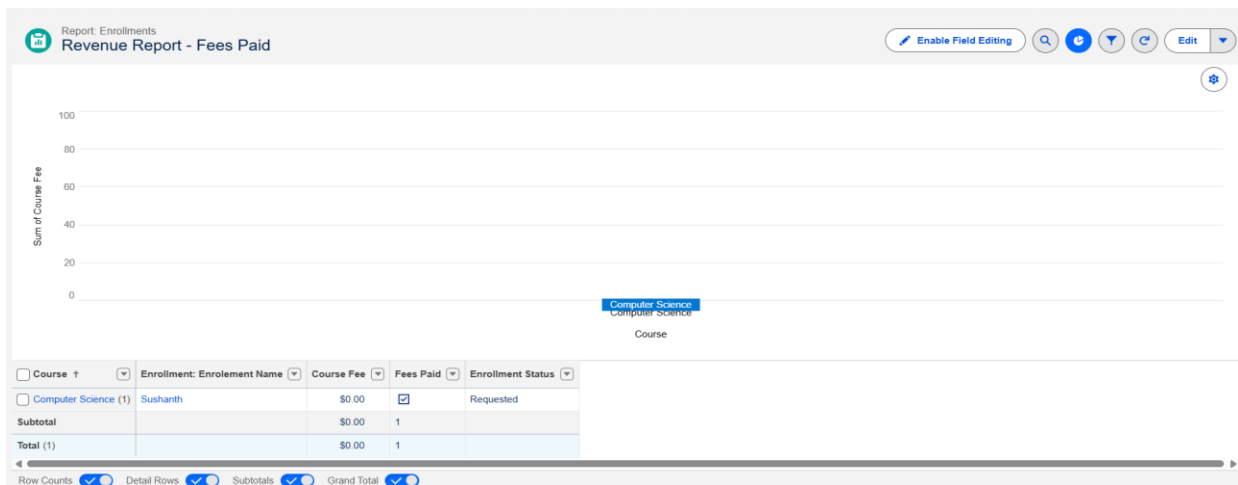


Activity 4: Create “Revenue Report (Fees Paid)” Report

- 1 . New Report → **Enrollments** report type.
- 2 . Filter: **Fees_Paid**_____c = TRUE.
- 3 . Group by **Course**_____r.Name.
- 4 . Summarize **Course_Fee**_____c or similar fee field.
- 5 . Save as **Revenue Report – Fees Paid**.

Activity 4: Create “Revenue Report (Fees Paid)” Report

- 6 . New Report → **Enrollments** report type.
- 7 . Filter: **Fees_Paid_____c = TRUE.**
- 8 . Group by **Course_____r.Name.**
- 9 . Summarize **Course_Fee_____c** or similar fee field
- 10 . Save as **Revenue Report – Fees Paid.**

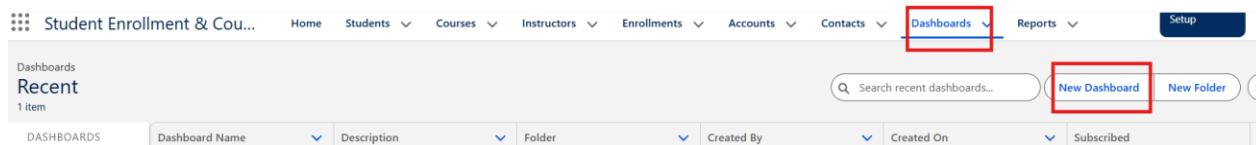


Milestone 12 – Dashboards:

Activity 1: Create “Student Managemet Dnashboard”

Steps:

- 1 . Go to **Dashboards** → **New Dashboard.**
- 2 . Name: **Student Management Dashboard.**



- 3 . Select folder → **Create.**
- 4 . Select new Dashboard folder and save.

Add components:

- Donut Chart** → using **Students by Status** report.
- Bar Chart** → using **Enrollments by Course** report

Gauge → using **Revenue**
revenue field).

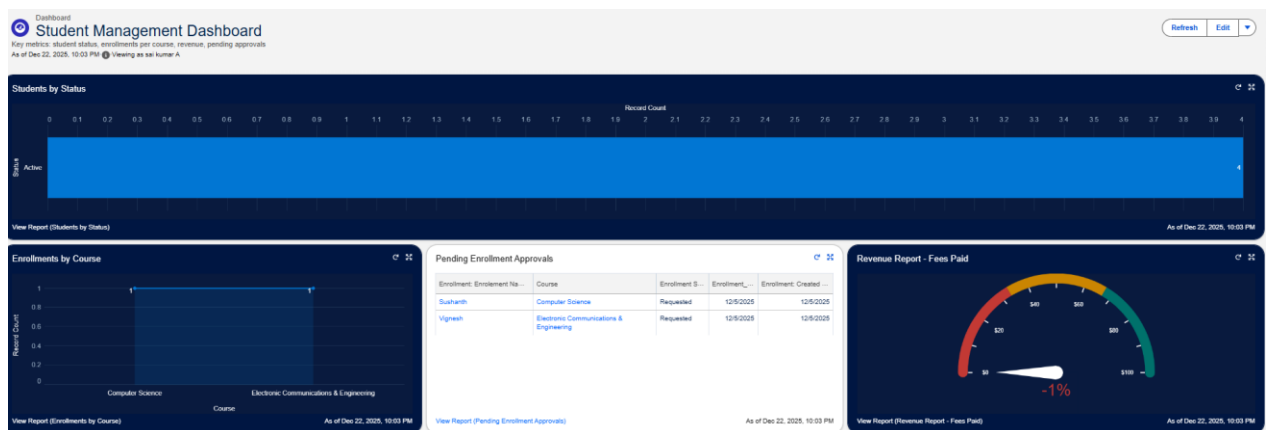
Report – Fees Paid (sum of

Table → using **Pending**

Enrollment Approvals report.

5 . Arrange components on canvas.

6 . Click **Save** → **Done**.



Milestone 13 – Security Setup

Activity 1: Configure Profiles

);

1 . Go to **Setup** → **Profiles** → **New** (or Clone **Profiles Required:**

Admin – Full access.

Enrollment Officer – Create/Update Enrollments.

Instructor – Read-only access to assigned Enrollments.

2 . **Steps (example for Enrollment Officer fName: Enrollment Officer.**

3 . In **Object Settings**:

○ **Enrollment_____c**: Read, Create, Edit.

○ **Student_____c / Course_____c**: Read (and Edit if required).

4 . Save.

Repeat similarly for **Instructor** profile with **Read-Only** on Enrollments.

Activity 2: Create Permission Sets

- 1 . Go to **Setup** → **Permission Sets** → **New**.
- 2 . Create **Custom Object Access** (additional access to project objects if needed).
- 3 . Create **Dashboard Editor** (Create/Customize Reports & Dashboards permissions).
- 4 . Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

- 1 . Go to **Setup** → **Sharing Settings**.
- 2 . Set **Organization-Wide Defaults**:
 - Students (Student__c)**: Public Read/Write.
 - Courses (Course__c)**: Public Read/Write.
 - Enrollments (Enrollment__c)**: Private.
- 3 . Save.

For Instructor access to assigned Enrollments, optionally:

Add a **User lookup** on Enrollment (Instructor User).

- 4 . Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

7.FUNCTIONAL & PERFORMANCE TESTING

Functional and performance testing for the Student Management & Course Management System (SECMS) ensures that all core features—such as user authentication, student enrollment, course creation and registration, attendance marking, grade entry, and reporting—work correctly and efficiently under real-world conditions. Functional testing uses structured test cases to validate each requirement, confirming that valid and invalid inputs are handled properly, role-based access is enforced, and business rules like prerequisites and seat limits are applied correctly. Performance testing complements this by simulating multiple concurrent users and large data volumes to verify that the system maintains acceptable response times, remains stable during peak enrollment periods, and scales without errors or data corruption, thereby ensuring a reliable and responsive platform for students, faculty, and administrators.

8.ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9.Conclusion:

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce’s automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10.FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11.APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You